IFROS RECOMMENDATION LETTER

Applicant's Name: Mostafa Ahmed Mostafa Ahmed.

Recommender's name: Ayman Nada.

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I am delighted to write this recommendation letter in support of Mostafa Ahmed Mostafa's application for the **Intelligent Field Robotic Systems (IFROS)** Master's program. As a professor at Egypt-Japan University of Science and Technology (E-JUST), I have had the pleasure of teaching Mostafa in several core courses in the Mechatronics and Robotics Engineering program, including **Automatic Control 1, Automatic Control 2**, and **Mechanical Vibrations**. Throughout these courses, Mostafa consistently stood out as an exceptionally driven and talented student, excelling in both theoretical and practical aspects of the material.

Mostafa's performance in these courses was exemplary, consistently ranking among the top students in his cohort. His deep understanding of control systems and his ability to apply theoretical concepts to complex, real-world mechatronic systems were evident in every aspect of his work. His analytical skills, critical thinking, and problem-solving abilities were particularly evident in his projects, where he demonstrated an aptitude for designing innovative and effective solutions. Mostafa's research in my lab on **FPGA-based mechatronic systems** further demonstrated his capability to engage with advanced topics in embedded systems and control, as well as his ability to execute high-quality research.

One of the most impressive projects Mostafa worked on under my supervision was his work on a **self-balancing robot**, where he analyzed and implemented a variety of control strategies, including **PID, PD**, and **lead-lag controllers**. This project required meticulous attention to detail, as even small variations in control parameters could significantly affect the system's performance. Mostafa's ability to fine-tune these controllers, optimize performance, and adapt to challenges showcased his strong engineering intuition and practical skills.

Mostafa also demonstrated excellent leadership and teamwork throughout his academic career. He led multiple projects, including his role in the **E-JUST Racing Team**, where he spearheaded the design and development of an electric vehicle prototype. His ability to organize tasks, collaborate effectively with peers, and manage both technical and non-technical aspects of the project was pivotal in the team's success. His technical leadership was also highlighted by his participation in various robotics and automotive competitions, where his contributions to both design and implementation were crucial to their success.

Beyond his academic and technical strengths, Mostafa is an exceptionally motivated and disciplined individual. He has shown a passion for robotics and automation that extends far beyond his coursework. His commitment to developing practical solutions in the field, as demonstrated through his extensive work on robotics projects, and his ongoing independent research, which includes several published papers, highlights his determination to make meaningful contributions to the field of mechatronics and robotics.

Mostafa's strong academic background, research experience, and leadership skills, combined with his passion for robotics, make him an ideal candidate for the **IFROS Master's program**. I am confident that Mostafa will excel in this program, contribute to the research community, and continue to grow as an expert in the field of robotics. I wholeheartedly recommend him for admission to the IFROS program and believe that he will bring valuable insights and contributions to the field of robotics.

Please do not hesitate to contact me if you require any additional information regarding Mostafa's qualifications.

Signature  Date: 14 March 2025